

Marvel universe through movies try and lecture them about the nuances of the storyline. But I digress.

Just as tattoos were once ink that adorned the bodies of the real rebels of societies, but eventually became so mainstream that it almost feels like everyone has at least one in the western world... many trends eventually become so mainstream that it's hard to believe there was a time they weren't accepted. Denim, for example, was invented for labourers in order to last longer under tough conditions, then went on to become a rebellious fashion statement, and now is so mainstream that no one even cares about it.

Bodyhacking or cybernetics is as far from mainstream as it gets, for now. Although it may not be very rebellious, it is still classified as weird. Most people don't like the idea of hacking into one's own skin to make modifications, even though most bodyhackers will point out that people said the same thing about tattoos or piercings in decades gone past, and yet they're far more normal and accepted now.

## BODYHACKING?

To be honest, people are weird, and they have hacked their bodies in all sorts of weird ways. There are people who think they're tigers, who have had surgeries and implants to look more like a tiger... someone else thinks they're the devil, and gets horns on the top of their head. Often, bodyhacking is seen as something the people bordering on clinically insane do. Essentially, it's just the act of modifying one's body in an artificial way, for whatever reason. If you're curious, Google is your friend. What we are going to focus on is the hacking of one's body with technology that improves the function of the body in some way. We don't care if it's weird or not, but we do care if it's useful or not.

## CYBERNETICS?

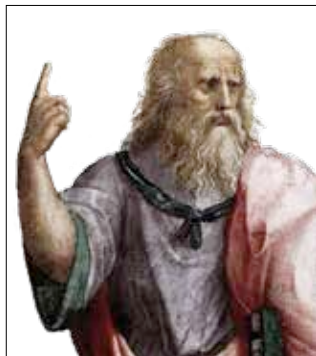
Before we progress any further we also need to explain what

we mean by cybernetics for the purpose of this cover theme. The problem with the word "cybernetics" is that it is used to describe so many different things that it's almost impossible to come up with a definition that everyone agrees on.

The closest might very well be the dictionary definition which states that it is the science of communication and automatic control systems in both machines and living things. However, that has to be a little vague on purpose, because it covers so many different disciplines. It's also been called "the science and art of the understanding of understanding", or "the study of systems and processes that interact with themselves and produce themselves from themselves", and also "a branch of mathematics dealing with problems of control, recursiveness, and information, focuses on forms and the patterns that connect."

Confused? Don't worry, you're in good company. Because the term cybernetics is used in mathematics, biology, computer science, sociology, art, architecture, law, sport, management, psychology, etc., it is very hard to arrive at a concrete definition, which is why we're inventing one for the purpose of this bunch of articles.

When we say cybernetics, we mean it in a purely medical and artificial intelligence way, specifically towards the goal of becoming cyborgs. Medical implants that work without outside intervention, and naturally, just as the original body part would... that's essentially cybernetics for us within the scope of this cover story.



Plato was the first to use the term cybernetics, but he meant "the study of self-governance"

## CYBORGS

So what does it take to become or be termed a cyborg? Instead of throwing about more definitions let's look at examples and go from there.

### Kevin Warwick

If you've been reading Digit for long enough, you will remember that we first introduced you to Kevin

Warwick way back in January of 2005. He even wrote a column for us in our fourth anniversary issue, where he outlined his vision for human-machine convergence. Thus, it seems appropriate to always start all conversations about Cyborgs with this mild-mannered teacher. He was Professor of Cybernetics back then, and is now Deputy Vice-Chancellor (Research) at Coventry University. Kevin gave himself the title, the world's first Cyborg because he implanted a chip in his hand as early as August 1998. That chip allowed computers to track his movement through the halls and rooms of the Department of Cybernetics in the University of Reading. He



Kevin Warwick – the world's first cyborg

also had an electronic array inserted into his arm, which was then able to control a robot arm – essentially mimicking the movement of his own hand. Later, they even implanted a simpler sensor into his wife's arm, and tried to connect them both together electronically. When Warwick moved his arm, his wife felt like she was moving her arm. She wasn't. It was the first time that the nervous systems of two different humans had communicated directly. A possibility that would have been interesting was if such implants would allow one person to actually move another person's body – which would bring an all new meaning to the term bodyhacking!

### Jesse Sullivan

In 2001, while working as an electrician in Tennessee, US, and repairing power lines for the power company he worked for, Jesse accidentally touched a live wire that was carrying over